15.081 Mathematical programming, HW3

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1. **Exercise 4.4: (Given)** A symmetric, square matrix. LP $\min_{Ax \geq c, x \geq 0} c^T x$ **(Prove)** If x^* satisfies $Ax = c, x \geq 0$ then x^* is an optimal solution

   Proof: We begin by formulating the primal and dual problems:

   $$(P): \min_x \quad c^T x \qquad\qquad (D): \max_p \quad p^T c$$
   $$\text{s.t.} \quad Ax \geq c \qquad\qquad \text{s.t.} \quad p \geq 0$$
   $$x \geq 0 \qquad\qquad p^T A \leq c$$

   We are given x^* such that $Ax^* = c, x^* \geq 0$. This x^* is feasible for the primal problem, (P).

   Now, A symmetric $\implies Ax = x^T A$ so $p = x^*$ is also feasible for the dual problem, (D):

   - $p = x^* \geq 0$
   - $p^T A = (x^*)^T A = Ax = c$

   What's more, the objective values of the primal and dual problems are equal: $c^T x^* = c^T p$ for $p = x^*$. Together, i) x^* primal feasible, ii) $p = x^*$ dual feasible and iii) equal objectives prove that x^* is optimal for the primal problem by weak duality (Corrolary 4.2 in book)

2. **Exercise 4.10: Saddlepoints of the Lagrangian**
   (Given) Standard form problem $\min_{Ax=b, x \geq 0} c^T x$ and define Lagrangian

   $$L(x, p) = c^T x + p^T (b - Ax)$$

   A pair $(x^*, p^*), x^* \geq 0$ is a saddlepoint if

   $$L(x^*, p) \leq L(x^*, p^*) \leq L(x, p^*) \quad \forall x \geq 0, p$$

   (Prove) (x^*, p^*) is a saddlepoint $\iff$ x^* and p^* are primal and dual optimal, respectively

   Proof: Begin by stating the primal and dual problems:

   $$(P): \min_x \quad c^T x \qquad\qquad (D): \max_p \quad p^T b$$
   $$\text{s.t.} \quad Ax = b \qquad\qquad \text{s.t.}$$
   $$x \geq 0 \qquad\qquad p^T A \leq c$$

(a) ($\Rightarrow$) Assume $(x^*, p^*), x^* \geq 0$ is a saddlepoint, then

i. (p^* dual feasible)

$$\begin{aligned} L(x^*, p^*) &\leq L(x, p^*) \quad \forall x \geq 0 \\ c^T x^* + p^{*T}(b - Ax^*) &\leq c^T x + p^{*T}(b - Ax) \quad \forall x \geq 0 \\ c^T x^* - p^{*T} A x^* &\leq c^T x - p^{*T} A x \quad \forall x \geq 0 \\ (c^T - p^{*T} A)(x^* - x) &\leq 0 \quad \forall x \geq 0 \end{aligned}$$

We note that $(c^T - p^{*T} A) \geq 0$ by contradiction: Consider this inequality componentwise

$$(c_i - p^{*T} A_i)(x_i^* - x_i) \leq 0 \quad \forall x_i \geq 0$$

and suppose $(c_i - p^{*T} A_i) < 0$, then we can pick feasible $x_i > x_i^* \geq 0$ which would yield a contradiction:

$$\underbrace{(c_i - p^{*T} A_i)}_{<0} \underbrace{(x_i^* - x_i)}_{<0} > 0$$

and hence $(c^T - p^{*T} A) \geq 0 \Longrightarrow p^{*T} A \leq c^T \Longrightarrow p^*$ dual feasible

ii. (x^* primal feasible)

$$\begin{aligned} L(x^*, p) &\leq L(x^*, p^*) \quad \forall p \\ c^T x^* + p^T(b - Ax^*) &\leq c^T x^* + p^{*T}(b - Ax^*) \quad \forall p \\ (p - p^*)^T(b - Ax^*) &\leq 0 \quad \forall p \end{aligned}$$

By a similar argument as above, we note that $(b - Ax^*) = 0$ otherwise we could pick a feasible p_i such that $(p_i - p_i^*)^T(b_i - A_i x^*) > 0$ (or < 0) yielding a contradiction. Hence $Ax^* = b,\ x^* \geq 0 \Longrightarrow x^*$ primal feasible

iii. (x^* primal optimal, p^* dual optimal)
Returning to the saddlepoint inequalities

$$\begin{aligned} L(x^*, p^*) &\leq L(x, p^*) \quad \forall x \geq 0 \\ c^T x^* + p^{*T}(b - Ax^*) &\leq c^T x + p^{*T}(b - Ax) \quad \forall x \geq 0 \\ c^T x^* &\leq p^{*T} b + (c^T - p^{*T} A)x \quad \forall x \geq 0, Ax^* = b \\ c^T x^* &\leq p^{*T} b \quad \text{for } x = 0, Ax^* = b \end{aligned}$$

x^* and p^* primal and dual feasible $\Rightarrow p^{*T} b \leq c^T x^*$ by weak duality, hence $c^T x^* = p^{*T} b$. By strong duality, x^* and p^* are primal and dual optimal.

(b) ($\Leftarrow$) Assume x^* and p^* are primal and dual optimal, respectively

i. **Show:** $L(x^*, p^*) \leq L(x, p^*) \quad \forall x \geq 0$

$$\begin{aligned}
c^T x^* &= p^{*T} b && \text{(strong duality)} \\
c^T x^* + p^{*T}(b - Ax^*) &= p^{*T} b && (x^* \text{ optimal} \Rightarrow (b - Ax^*) = 0) \\
c^T x^* + p^{*T}(b - Ax^*) &\leq p^{*T} b + (c^T - p^{*T} A) && (p^* \text{ optimal} \Rightarrow (c^T - p^{*T} A) \geq 0) \\
c^T x^* + p^{*T}(b - Ax^*) &\leq p^{*T} b + (c^T - p^{*T} A)x \quad \forall x \geq 0 \\
c^T x^* + p^{*T}(b - Ax^*) &\leq c^T x + p^{*T}(b - Ax) \quad \forall x \geq 0 \\
L(x^*, p^*) &\leq L(x, p^*) \quad \forall x \geq 0
\end{aligned}$$

ii. **Show:** $L(x^*, p) \leq L(x^*, p^*) \quad \forall p$ This is trivially true:

$$\begin{aligned}
p^T(b - Ax^*) &= p^{*T}(b - Ax^*) \quad \forall p && (x^* \text{ optimal} \Rightarrow (b - Ax^*) = 0) \\
c^T x^* + p^T(b - Ax^*) &= c^T x^* + p^{*T}(b - Ax^*) \\
L(x^*, p) &= L(x^*, p^*) \quad \forall p \\
L(x^*, p) &\leq L(x^*, p^*) \quad \forall p
\end{aligned}$$

3. **Exercise 4.20*: Strict complementary slackness**

(a) **(Given)** Primal problem $(P) : \min_{\{x|Ax=b, x\geq 0\}} c^T x$ and Dual problem $(D) : \max_{\{p|p^T A \leq c\}} p^T b$, both with optimal solutions. Every optimal solution to (P) has $x_j = 0$
(Show) $\exists$ optimal p for (D) such that $p^T A_j < c_j$

Let $(x*, p*)$ be optimal values for $(P), (D)$. Consider new perturbed problems $(P'), (D')$ with $\delta > 0$ sufficiently small

$$\begin{array}{lll}
(P') \min\limits_{x} & c^T x - \delta x_j & \\
\text{s.t.} & Ax = b & \\
& x \geq 0 &
\end{array}
\qquad\qquad
\begin{array}{lll}
(D') \max\limits_{p} & p^T b & \\
\text{s.t} & p \text{ free} & \\
& p^T A \leq c^T - \delta e_j &
\end{array}$$

Feasibility of $x*$ for (P'): For optimal BFS, B, $x* = B^{-1}b$, and is unaffected by the perturbation

Optimality of $x*$ for (P'): We claim that for sufficiently small, δ, the optimality condition holds. From the given, we know every optimal x^* for (P) has $x_j = 0$. Any feasible direction that could improve the value of (P') through δx_j therefore would reduce the value of $c^T x^*$. Simply choose δ such that $\bar{c}' = (c - \delta e_j) - c_B B^{-1} A \geq 0$. This is possible whenever $c_j - c_B B^{-1} A_j > \delta$ by considering the cost reduction vector for $(P'), \bar{c}'$:

$$\bar{c}' = (c - \delta e_j) - c_B B^{-1} A \Longrightarrow \delta \leq c_j - c_B B^{-1} A_j$$

Note, if for the given BFS, B, $c_j = c_B B^{-1} A_j$, we can simply pivot to another BFS where index j is nonbasic. This is possible since all optimal x^* have $x_j = 0$

Feasible p for (D'): Now by the existence of an optimal in (P'), we know (D') has a feasible solution. By construction of (D'), for any feasible p of (D'), we have

$$p^T A \leq c^T - \delta e_j \Longrightarrow \begin{cases} p^T A_i \leq c_i - \delta < c_i & \text{if } i = j \\ p^T A_i \leq c_i & \text{otherwise} \end{cases}$$

So all feasible p for (D') satisfy $p^T A_j < c_j$ and $p^T A_i \leq c_i, i \neq j$. This feasible set is contained within the feasible set of our original dual problem, (D).

Optimal p for (D'): By strong duality, we have existence of an optimal p for (D').

Connecting (D') to (D): By strong duality, feasible p of (D') is optimal if $b^T p = c^T x^*$. $c^T x^*$ is the same objective value as our original (D). Furthermore, the feasible set of (D') is contained within the feasible set of (D). Therefore, any p feasible in (D') is feasible in (D), so optimal p in (D') is also optimal in (D). So we've demonstrated that $\exists$ optimal p for (D) such that $p^T A_j < c_j$.

(b) **(Show)** There exist optimal solutions x, p to primal and dual problems, $(P), (D)$ respectively such that $\forall j$ we have either $x_j > 0$ or $p^T A_j < c_j$
(Hint) Use part (a) for each j, and then take the average of the vectors obtained
(Proof) Consider the sets of optimal solutions and corresponding sets of indices:

$$\mathcal{X} = \{x \mid Ax = b, x \geq 0, c^T x = val(P)\} \qquad \mathcal{P} = \{p \mid p^T A \leq c, p^T b = val(D)\}$$
$$S = \{j \mid \exists x \in \mathcal{X}, x_j > 0\} \qquad S^C = \{j \mid \exists x_j = 0 \; \forall x \in \mathcal{X}\}$$

From given we know $\exists x \in \mathcal{X}$. We construct a primal optimal solution as the average of $|S|$ primal optimal solutions. Let $x^{(j)}$ be $x \in \mathcal{X}$ with $x_j^{(j)} > 0$. Then let

$$\bar{x} = \frac{1}{|S|} \sum_{k \in S} x^{(k)}, \text{ noting}$$
$$\bar{x}_j = \frac{1}{|S|} \sum_{k \in S} x_j^{(k)} \geq \frac{x_j^{(j)}}{|S|} > 0 \quad \forall j \in S$$
$$c' \bar{x} = \frac{1}{|S|} \sum_{j \in S} c' x^{(j)} = \frac{|S|}{|S|} val(P) = val(P) \text{ so } \bar{x} \in \mathcal{X}$$

We also consider a dual optimal solution as the average of $|S^C|$ dual optimal solu-

tions. Using part (a), for any $j \in S^C$ we can let $p^{(j)}$ be $p \in \mathcal{P}$ with $p^{(j)T} A_j < c_j$.

$$\bar{p} = \frac{1}{|S^C|} \sum_{k \in S^C} p^{(k)}, \text{ noting}$$

$$\bar{p}^T A_j = \frac{1}{|S^C|} \sum_{k \in S^C} p^{(k)T} A_j = \frac{1}{|S^C|} \left(p^{(j)T} A_j + \sum_{k \neq j} p^{(k)T} A_j \right) < \frac{1}{|S^C|} \left(c_j + \sum_{k \neq j} c_j \right) = c_j \quad \forall j \in S^C$$

$$\bar{p}^T b = \frac{1}{|S^C|} \sum_{k \in S^C} p^{(k)T} b = \frac{|S^C|}{|S^C|} val(D) = val(D) \text{ so } \bar{p} \in \mathcal{P}$$

Now we have new primal dual optimal solutions $\bar{x}, \bar{p}$. Now since sets S, S^C partition the index set, we have $\forall j \in [n]$ either $x_j > 0$ or $p^T A_j < c_j$.

(c) **(Given)** Primal problem $(P) : \min_{\{x | Ax \geq b, x \geq 0\}} c^T x$ and Dual problem $(D) : \max_{\{p | p^T A \leq c, p \geq 0\}} p^T b$, both with optimal solutions.
(Show) There exist optimal solutions to $(P), (D)$ that satisfy strict complementary slackness. That is,

- $\forall\, j$ either $x_j > 0$ or $p^T A_j < c_j$
- $\forall\, i$ either $p_i > 0$ or $a_i^T x > b_i$

(Proof) Following the hint, we convert (P) to an equivalent standard form (P'):

$$\begin{array}{lllll} (P') : \min\limits_{x} & c^T x & \Longrightarrow & \min\limits_{x} & \bar{c}^T \bar{x} \qquad (\bar{c} = [c, 0] \in \mathbb{R}^{n+m}) \\ \text{s.t.} & [A, -I][x, s]^T = b & & \text{s.t.} & \bar{A}\bar{x} = b \quad (\bar{A} = [A, -I] \in \mathbb{R}^{m \times (n+m)}) \\ & x \geq 0, s \geq 0 & & & \bar{x} \geq 0 \qquad (\bar{x} = [x, s] \in \mathbb{R}^{n+m}) \end{array}$$

And using the lagrangian,

$$g(p) = \min_{\bar{x} \geq 0} L(\bar{x}, p) = \min_{\bar{x} \geq 0} \bar{c}^T \bar{x} + p^T (b - \bar{A}\bar{x}) = \bar{p}^T b + \min_{\bar{x} \geq 0} (\bar{c}^T - \bar{p}^T \bar{A}) \bar{x}$$

We formulate the dual problem

$$\begin{array}{lllll} (D') : \max\limits_{p} & p^T b & \Longleftrightarrow & (D) : \max\limits_{p} & p^T b \\ \text{s.t.} & p \text{ free} & & \text{s.t.} & p \text{ free} \\ & p^T [A, -I] \leq [c, 0]^T & & & p^T A \leq c^T, p \geq 0 \end{array}$$

We have shown that (P) is equivalent to (P') in standard form, who's dual, (D'), is equivalent to (D). By part (b), we know that there exist optimal solutions $\bar{x}, p$ to $(P'), (D')$ that satisfy strict complementary slackness. The optimal p is the same for $(D), (D')$ and the corresponding x for (P) is just the first n components of $\bar{x}$.

4. **Exercise 4.27: (Given)** Matrix A **(Show)** the following are equivalent (iff):

    1. Any $x \in S = \{x \mid Ax \geq 0, x \geq 0\}$ satisfies $x_1 = 0$

2. $\exists\, p \geq 0$ such that $p^T A \leq 0,\ p^T A_1 < 0$

(a) $(1 \Rightarrow 2)$ Consider the following primal and dual pair of optimization problems

$$(P):\ \min_x \quad -e_1^T x \qquad\qquad (D):\ \max_p \quad 0$$
$$\text{s.t.} \quad Ax \geq 0 \qquad\qquad \text{s.t.} \quad p \geq 0$$
$$x \geq 0 \qquad\qquad p^T A \leq -e_1^T$$

From the givens, we know the only feasible x for (P) are such that $-e_1^T x = 0$. By strong duality, this means we also achieve an optimum in our dual problem, (D), with feasible p. Examining such a p we see:

$$\begin{matrix} p \geq 0 \\ p^T A \leq -e_1^T \end{matrix} \Longrightarrow \begin{cases} p^T A_1 \leq -1 \\ p^T A_{i>1} \leq 0 \end{cases} \Longrightarrow \begin{cases} p \geq 0 \\ p^T A_1 < 0 \\ p^T A \leq 0 \end{cases}$$

These are precisely the p we wanted to prove existed.

(b) $(2 \Rightarrow 1)$ Start with any $x \geq 0$ such that $Ax \geq 0$. Left-multiply by $p^T \geq 0$:

$$0 \leq p^T(Ax) = \sum_i p^T A_i x_i = p^T A_1 x_1 + \sum_{i>1} p^T A_i x_i$$
$$0 \leq p^T A_1 x_1 + \sum_{i>1} 0 * x_i \qquad \text{(since } p^T A_i \leq 0\text{)}$$

So $0 \leq p^T A_1 x_1$ but $p^T A_1 < 0 \Longrightarrow x_1 = 0$ for any $x \geq 0$.

5. **Exercise 4.40: Extreme rays of a cone are extreme points of its sections**
(Given) Cone $C = \{x \in \mathbb{R}^n \mid a_i^T x \geq 0, i \in [m]\}$ with the first n constraint vectors linearly independent. For any nonnegative $r \in \mathbb{R}$, define polehydron

$$P_r = \{x \in C \mid \sum_{i=1}^n a_i^T x = r\}$$

(a) **(Show)** P_r is a bounded polyhedron for any $r \geq 0$
(Contradiction) Assume P_r is unbounded $\Longrightarrow \exists\, d \neq 0$ such that $x + \lambda d \in P_r\ \forall \lambda \geq 0$. Then we have

$$x + \lambda d \in C: \quad a_i^T(x + \lambda d) \geq 0 \Longrightarrow a_i^T d \geq 0 \qquad \forall i \in [m]$$
$$x + \lambda d \in P_r: \quad \sum_{i=1}^n a_i^T(x + \lambda d) = r \Longrightarrow r + \lambda \sum_{i=1}^n a_i^T d = r \Longrightarrow \sum_{i=1}^n a_i^T d = 0$$

The only way we have these two conditions together is for $a_i^T d = 0\ \forall i \in [n]$. And by the linear independence of the first n constraint vectors, this implies that $d = 0$, which contradicts our assumption. Hence P_r is bounded.

(b) **(Show)** For $r > 0, x \in P_r$ is an extreme point of $P_r \iff x$ is an extreme ray of cone C

(Definition) Extreme ray of a cone: $x \neq 0 \in C$ is an extreme ray if there are n - 1 linearly independent constraints that are active at x

i. ($\Longrightarrow$) x an extreme point of $P_r \implies n$ linearly independent constraints of P_r active at x (since extreme point $\Rightarrow$ basic solution). By construction, one of these active constraints must be $\sum_{i=1}^{n} a_i^T x = r$. The other $n-1$ linearly independent active constraints must come from the requirement that $x \in C$. Therefore, there are $n-1$ linearly independent constraints in C that are active at x, proving x is an extreme ray of cone C.

ii. ($\Longleftarrow$) x an extreme ray of cone $C \implies$ there are $n-1$ linearly independent constraints in C that are active at x. By construction, of P_r, x has $n-1$ linearly independent constraints from requirement that $x \in C$, and one from the requirement that $\sum_{i=1}^{n} a_i^T x = r$. Therefore, x has n linearly independent constraints in P_r and it is a BFS $\Rightarrow x$ is an extreme point of P_r.